

# Power Analysis

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## Contents

|                                                                 |          |
|-----------------------------------------------------------------|----------|
| <b>1. Introduction and Effect Size Justification</b>            | <b>2</b> |
| Research Hypothesis . . . . .                                   | 2        |
| Data Generating Functions . . . . .                             | 2        |
| Simulation Scenarios . . . . .                                  | 2        |
| Scenario A: Conservative Estimate (Between-Subjects) . . . . .  | 2        |
| Scenario B: B: Optimistic Estimate (Between-Subjects) . . . . . | 2        |
| Scenario C: Crossover Design (Within-Subjects) . . . . .        | 2        |
| <b>2. The Data Generating Functions</b>                         | <b>3</b> |
| Scenario A: Conservative Estimate (Between-Subjects) . . . . .  | 3        |
| Scenario B: Optimistic Estimate (Between-Subjects) . . . . .    | 4        |
| Scenario C: Crossover Design (Within-Subjects) . . . . .        | 4        |
| <b>3. Power Estimation</b>                                      | <b>5</b> |
| <b>4. Visualization</b>                                         | <b>6</b> |
| Power Curve for each of the 3 scenarios . . . . .               | 6        |
| <b>5. Conclusion</b>                                            | <b>7</b> |
| <b>6. References</b>                                            | <b>7</b> |

# 1. Introduction and Effect Size Justification

## Research Hypothesis

We hypothesize that listening to **High-Tempo** ( $\geq 125$  BPM) music will significantly extend the **time-to-volitional-exhaustion (TTE)** during an isometric plank hold compared to listening to **Low-Tempo** ( $\leq 80$  BPM) music. This hypothesis draws on the theory of **psychological dissociation**, where fast-paced auditory stimuli compete with interoceptive fatigue signals (e.g., muscle burning), thereby lowering the **Rate of Perceived Exertion (RPE)** and delaying voluntary cessation of the exercise.

## Data Generating Functions

To ensure our power analysis simulation is grounded in reality, we derived our baseline parameters from recent literature specifically examining music and isometric strength tasks.

- **Baseline Performance ( $\mu$ ):** We assume a baseline plank hold time of **111 seconds**. This is based on the control group performance reported by *Feiss et al. (2021)* during a no-music plank condition.
- **Population Variance ( $\sigma$ ):** A critical challenge in measuring isometric endurance is the substantial natural variability in human strength. *Feiss et al. (2021)* reported a standard deviation of **43 seconds**, while *Poon et al. (2024)* reported SDs ranging from **40.9 to 58.3 seconds**. For our simulation, we will use **SD = 43 seconds**. This high variance (**CV around 39%**) suggests that individual strength differences may easily drown out the music effect in a **between-subjects** design.

## Simulation Scenarios

We will estimate statistical power under three distinct scenarios representing the range of effects reported in the literature.

### Scenario A: Conservative Estimate (Between-Subjects)

- **Basis:** *Feiss et al. (2021)* and *Centala et al. (2020)*.
- **Justification:** *Feiss et al.* found no significant difference in plank TTE between fast-tempo and no-music conditions. Similarly, *Centala et al.* found that while fast music improved the neuromuscular fatigue threshold, it did not significantly alter RPE at the point of exhaustion. This scenario assumes the null hypothesis is nearly true or the effect is negligible.
- **Effect Size:** **+5 seconds** (approx. Cohen's  $d = 0.12$ ).

### Scenario B: B: Optimistic Estimate (Between-Subjects)

- **Basis:** *Terry et al. (2020)* meta-analysis.
- **Justification:** Across **139 studies** and approximately **3,600 participants**, the generalized effect of music on physical performance was **Hedges'  $g = 0.31$** . This represents the average benefit one might expect from music across broad exercise contexts.
- **Effect Size:** **+13 seconds** ( $0.31 \times 43\text{s SD}$ ).

### Scenario C: Crossover Design (Within-Subjects)

- **Basis:** *Poon et al. (2024)*.
- **Justification:** In a study design very similar to ours (isometric plank), *Poon et al.* found that listening to music extended plank time from **144.8 s** to **171.0 s**, a difference of approximately **26 seconds**.

They reported a large effect size ( $d = 0.76$ ).

- **Design Note:** Crucially, *Poon et al.* utilized a **within-subjects (crossover)** design. To test whether we can replicate this power, Scenario C will simulate the same large effect (**+26 seconds**) but will use a **within-subjects DGP** in the code to demonstrate how controlling for individual baseline strength reduces error variance and increases power.

## 2. The Data Generating Functions

We define a set of functions `run_sim(scenario)` to simulate one iteration of the experiment under that scenario. This function generates potential outcomes, assigns treatment, and calculates the p-value.

### Scenario A: Conservative Estimate (Between-Subjects)

- Design: Randomized Controlled Trial (Between-Subjects).
- Effect Size: 5 seconds (Small effect, based on null findings in Feiss et al.).
- Logic: Uses Linear Regression with Robust Standard Errors (HC3) to handle high variance.

```
run_sim_scenario_A <- function(N) {  
  # 1. Define Parameters  
  baseline_mean <- 111 # Seconds (Feiss et al. Control Group)  
  population_sd <- 43  # Seconds (Feiss et al. SD)  
  effect_size   <- 5   # Conservative estimate (Small effect)  
  
  # 2. Generate Data (Between-Subjects)  
  d <- data.table(id = 1:N)  
  
  # Generate potential outcomes for Control (Y0) and Treatment (Y1)  
  d[, Y0 := rnorm(N, mean = baseline_mean, sd = population_sd)]  
  d[, Y1 := Y0 + effect_size]  
  
  # Random Assignment (50/50 split)  
  d[, T_assigned := 0]  
  treated_indices <- sample(1:N, size = N/2, replace = FALSE)  
  d[treated_indices, T_assigned := 1]  
  
  # Reveal Observed Outcomes  
  d[, Y_obs := T_assigned * Y1 + (1 - T_assigned) * Y0]  
  
  # 3. Conduct Analysis  
  mod <- lm(Y_obs ~ T_assigned, data = d)  
  
  # Calculate Robust Confidence Intervals  
  plank_robust_ci <- coefci(mod, vcov = vcovHC(mod, type = "HC3"), level = 0.95)  
  
  # Check significance: Does the CI for T_assigned exclude 0?  
  lower_bound <- plank_robust_ci["T_assigned", 1]  
  upper_bound <- plank_robust_ci["T_assigned", 2]  
  
  # Return 1 if significant (0 is outside the interval), else 0  
  is_significant <- as.numeric(!(lower_bound <= 0 && upper_bound >= 0))  
  return(is_significant)  
}
```

## Scenario B: Optimistic Estimate (Between-Subjects)

- Design: Randomized Controlled Trial (Between-Subjects).
- Effect Size: 26 seconds (Large effect, based on Poon et al.).
- Logic: Uses Linear Regression with Robust Standard Errors (HC3).

```
run_sim_scenario_B <- function(N) {  
  # 1. Define Parameters  
  baseline_mean <- 111 # Seconds  
  population_sd <- 43 # Seconds  
  effect_size <- 26 # Optimistic estimate (Based on Poon et al.)  
  
  # 2. Generate Data (Between-Subjects)  
  d <- data.table(id = 1:N)  
  
  # Generate potential outcomes  
  d[, Y0 := rnorm(N, mean = baseline_mean, sd = population_sd)]  
  d[, Y1 := Y0 + effect_size]  
  
  # Random Assignment (50/50 split)  
  d[, T_assigned := 0]  
  treated_indices <- sample(1:N, size = N/2, replace = FALSE)  
  d[treated_indices, T_assigned := 1]  
  
  # Reveal Observed Outcomes  
  d[, Y_obs := T_assigned * Y1 + (1 - T_assigned) * Y0]  
  
  # 3. Conduct Analysis (Linear Model with HC3 Robust SE)  
  mod <- lm(Y_obs ~ T_assigned, data = d)  
  
  # Calculate Robust Confidence Intervals  
  plank_robust_ci <- coefci(mod, vcov = vcovHC(mod, type = "HC3"), level = 0.95)  
  
  # Check significance  
  lower_bound <- plank_robust_ci["T_assigned", 1]  
  upper_bound <- plank_robust_ci["T_assigned", 2]  
  
  is_significant <- as.numeric(!(lower_bound <= 0 && upper_bound >= 0))  
  return(is_significant)  
}
```

## Scenario C: Crossover Design (Within-Subjects)

- Design: Crossover / Within-Subjects (Every participant does both).
- Effect Size: 26 seconds (Same large effect as Scenario B).
- Logic: Uses a Paired T-Test. This design controls for the large individual variability in plank time (the 43s SD), drastically increasing power.

```
run_sim_scenario_C <- function(N) {  
  # 1. Define Parameters  
  baseline_mean <- 111  
  population_sd <- 43  
  effect_size <- 26 # Optimistic estimate
```

```

# 2. Generate Data (Within-Subjects / Crossover)
# Generate N subjects, each with a specific "baseline strength"
subject_effect <- rnorm(N, mean = baseline_mean, sd = population_sd)

# Within-person variability (inconsistency between days)
# Usually smaller than population_sd. We estimate it at ~20s.
within_sd <- 20

# Condition 0: Low Tempo (Subject Baseline + Random Noise)
Y0 <- subject_effect + rnorm(N, mean = 0, sd = within_sd)

# Condition 1: High Tempo (Subject Baseline + Effect + Random Noise)
Y1 <- subject_effect + effect_size + rnorm(N, mean = 0, sd = within_sd)

# 3. Conduct Analysis (Paired T-Test)
# We test if the difference between Y1 and Y0 is significantly different from 0
test_result <- t.test(Y1, Y0, paired = TRUE)

# Return 1 if p < 0.05, else 0
return(as.numeric(test_result$p.value < 0.05))
}

```

### 3. Power Estimation

We now estimate statistical power by running each scenario's simulation **1,000 times** at a range of sample sizes. Power is calculated as the proportion of simulations that correctly reject the null hypothesis at the  $\alpha = 0.05$  level.

```

sample_sizes <- seq(20, 300, by = 20)
n_simulations <- 1000

power_A <- sapply(sample_sizes, function(n) {
  mean(replicate(n_simulations, run_sim_scenario_A(n)))
})

power_B <- sapply(sample_sizes, function(n) {
  mean(replicate(n_simulations, run_sim_scenario_B(n)))
})

power_C <- sapply(sample_sizes, function(n) {
  mean(replicate(n_simulations, run_sim_scenario_C(n)))
})

power_results <- data.table(
  N = rep(sample_sizes, 3),
  Power = c(power_A, power_B, power_C),
  Scenario = rep(c("A: Conservative (Between, +5s)",
                   "B: Optimistic (Between, +26s)",
                   "C: Crossover (Within, +26s)"), each = length(sample_sizes))
)

power_wide <- dcast(power_results, N ~ Scenario, value.var = "Power")
kable(power_wide, digits = 3, caption = "Estimated Power by Sample Size and Scenario")

```

Table 1: Estimated Power by Sample Size and Scenario

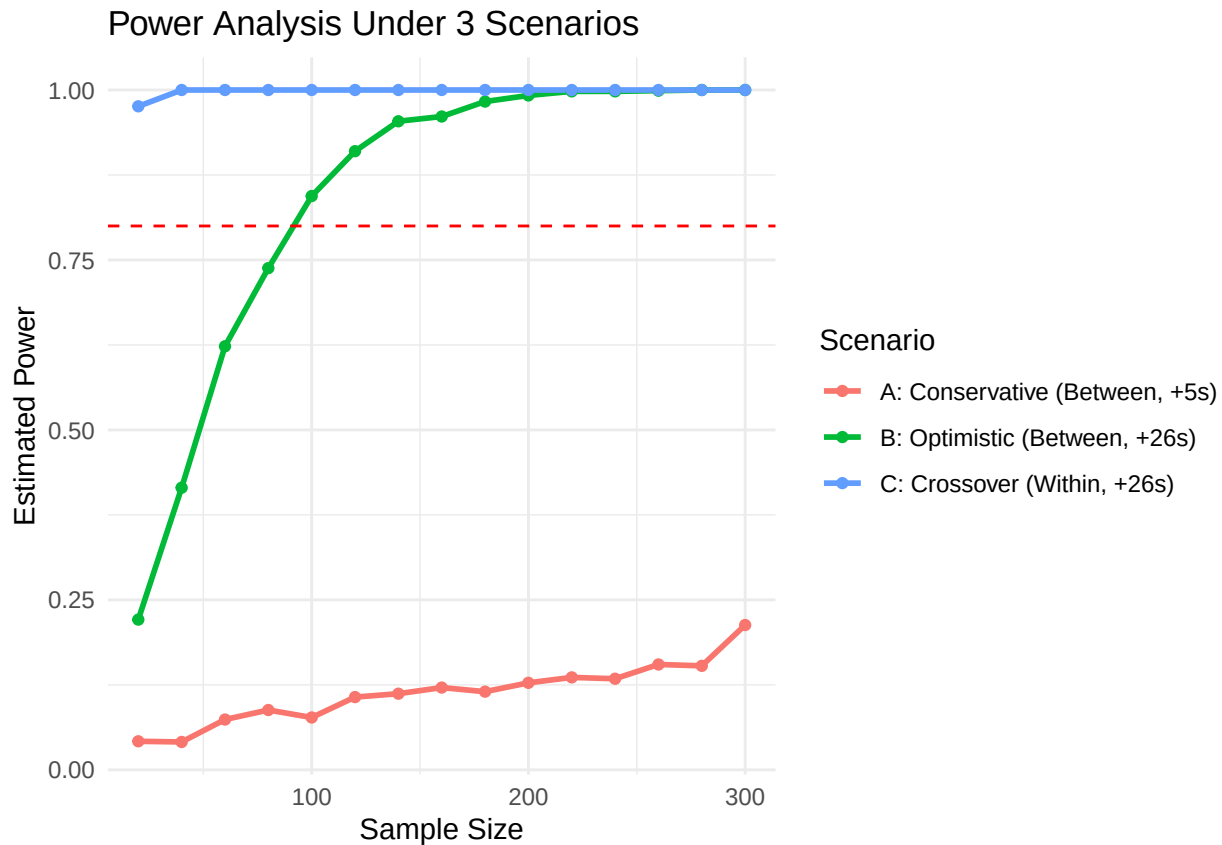
| N   | A: Conservative (Between, +5s) | B: Optimistic (Between, +26s) | C: Crossover (Within, +26s) |
|-----|--------------------------------|-------------------------------|-----------------------------|
| 20  | 0.042                          | 0.221                         | 0.976                       |
| 40  | 0.041                          | 0.415                         | 1.000                       |
| 60  | 0.074                          | 0.623                         | 1.000                       |
| 80  | 0.088                          | 0.738                         | 1.000                       |
| 100 | 0.077                          | 0.844                         | 1.000                       |
| 120 | 0.107                          | 0.910                         | 1.000                       |
| 140 | 0.112                          | 0.954                         | 1.000                       |
| 160 | 0.121                          | 0.961                         | 1.000                       |
| 180 | 0.115                          | 0.983                         | 1.000                       |
| 200 | 0.128                          | 0.992                         | 1.000                       |
| 220 | 0.136                          | 0.998                         | 1.000                       |
| 240 | 0.134                          | 0.998                         | 1.000                       |
| 260 | 0.155                          | 0.999                         | 1.000                       |
| 280 | 0.153                          | 1.000                         | 1.000                       |
| 300 | 0.213                          | 1.000                         | 1.000                       |

## 4. Visualization

### Power Curve for each of the 3 scenarios

```
ggplot(data = power_results) +
  aes(x=N, y=Power, color=Scenario) +
  geom_line(size=1) +
  geom_point() +
  geom_hline(yintercept=0.8, linetype='dashed', color='red') +
  labs(
    title = "Power Analysis Under 3 Scenarios",
    x = "Sample Size",
    y = "Estimated Power"
  ) +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



*Interpretation:* Under the conservative scenario (effect size = +5s), the design remains severely under powered, failing to exceed 0.20 even as the sample size approaches  $N = 300$ . The Optimistic Scenario (Between-Subjects) crosses the standard 0.80 power threshold at a sample size of approximately  $N \approx 90$ . The optimistic Crossover/Within-Subjects scenario is the most statistically efficient, achieving near-perfect power ( $>0.95$ ) with as few as 20 subjects.

## 5. Conclusion

In conclusion, the simulation demonstrates that the choice of experimental design is critical for successfully detecting an effect. The Conservative Scenario (A) proved unfeasible, failing to reach adequate power ( $<0.20$ ) even at maximal sample sizes. While the Between-Subjects design (Scenario B) is a valid option, it requires a substantial sample size of  $N \approx 90$  to reach the 80% power threshold. Consequently, Scenario C (Within-Subjects) emerges as the most promising design, achieving near-perfect power with as few as 20 subjects. Although within-subjects designs often carry the risk of carryover effects, we believe this threat is minimal for an isometric plank task performed only twice (low vs. high tempo). Therefore, given its superior feasibility and statistical efficiency, Scenario C is the group's preferred experimental design.

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